YZV211 Term Project

Relationship Analysis Between Electricity Generation, Greenhouse Gas Emissions and Deaths Due to Respiratory Diseases

In [35]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
```

In [36]:
```
df_e = pd.read_excel('enerji_kaynakları.xls', engine='xlrd')
df_g = pd.read_excel('sera gazi emisyonlari (milyon ton co2 esdegeri).xls', engine='xlrd')
df_d = pd.read_excel('ölüm_nedenleri.xls', engine='xlrd')
```

EDA (Exploratory Data Analysis)

Energy

In [37]:
```
df_e.head(10)
```

Out[37]:

	Enerji kaynaklarına göre elektrik enerjisi üretimi ve payları	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8	Unnamed: 9	Unnamed: 10
0	Electricity generation and shares by energy re...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	Yıl	NaN	Toplam	NaN	Kömür	Sıvı yakıtlar	Doğal gaz	Hidrolik	Yenilenebilir Enerji ve Atıklar (1)	NaN	NaN
2	Year	NaN	Total	NaN	Coal	Liquid fuels	Natural Gas	Hydro	Renewable Energy and wastes(1)	NaN	NaN
3	NaN	NaN	(GWh)	NaN	(%)	NaN	NaN	NaN	NaN	NaN	NaN
4	1970	NaN	8623	NaN	32.7	30.2	-	35.2	1.9	NaN	NaN
5	1971	NaN	9781	NaN	30.4	41.2	-	26.7	1.7	NaN	NaN
6	1972	NaN	11242	NaN	26	43.9	-	28.5	1.6	NaN	NaN
7	1973	NaN	12425	NaN	26.1	51.3	-	21	1.6	NaN	NaN
8	1974	NaN	13477	NaN	28.8	44.8	-	24.9	1.5	NaN	NaN
9	1975	NaN	15623	NaN	26.3	34.5	-	37.8	1.4	NaN	NaN

In [38]:
```
df_e_new = df_e.iloc[4:56,[0,2,4,5,6,7,8]] #Clean data
```

In [39]:
```
df_e_new.columns = ['Year','Total (GWh)','Coal%','Liquid Fuels%',
                    'Natural Gas%','Hydro%','Renewable Energy and Wastes%']
df_e_new.head()
```

Out[39]:

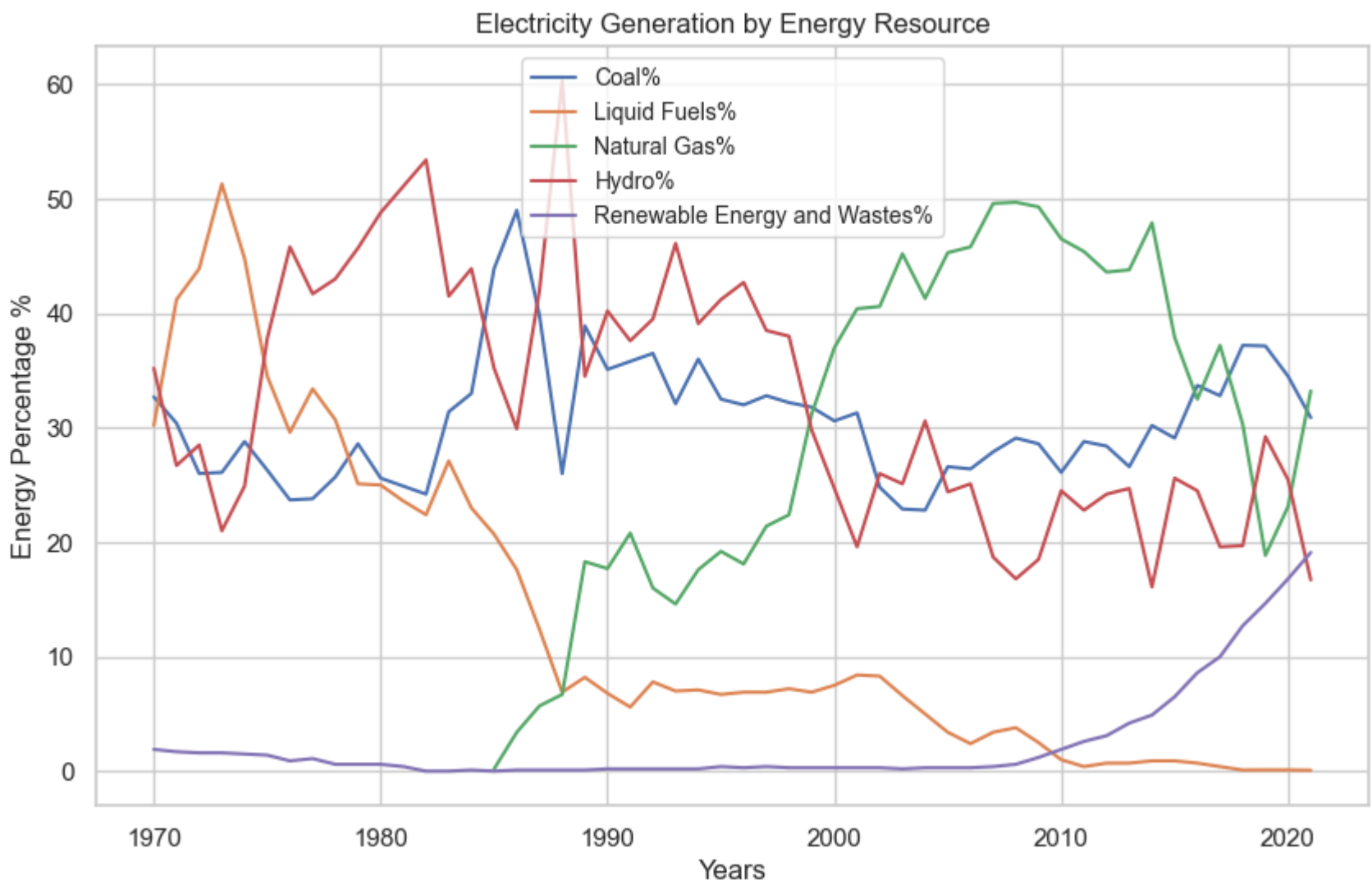
	Year	Total (GWh)	Coal%	Liquid Fuels%	Natural Gas%	Hydro%	Renewable Energy and Wastes%
4	1970	8623	32.7	30.2	-	35.2	1.9
5	1971	9781	30.4	41.2	-	26.7	1.7
6	1972	11242	26	43.9	-	28.5	1.6
7	1973	12425	26.1	51.3	-	21	1.6
8	1974	13477	28.8	44.8	-	24.9	1.5

In [40]:
```
df_e_new.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 52 entries, 4 to 55
Data columns (total 7 columns):
 #   Column                        Non-Null Count  Dtype 
---  ------                        --------------  ----- 
 0   Year                          52 non-null     object
 1   Total (GWh)                   52 non-null     object
 2   Coal%                         52 non-null     object
 3   Liquid Fuels%                 52 non-null     object
 4   Natural Gas%                  52 non-null     object
 5   Hydro%                        52 non-null     object
 6   Renewable Energy and Wastes%  52 non-null     object
dtypes: object(7)
memory usage: 3.0+ KB
```

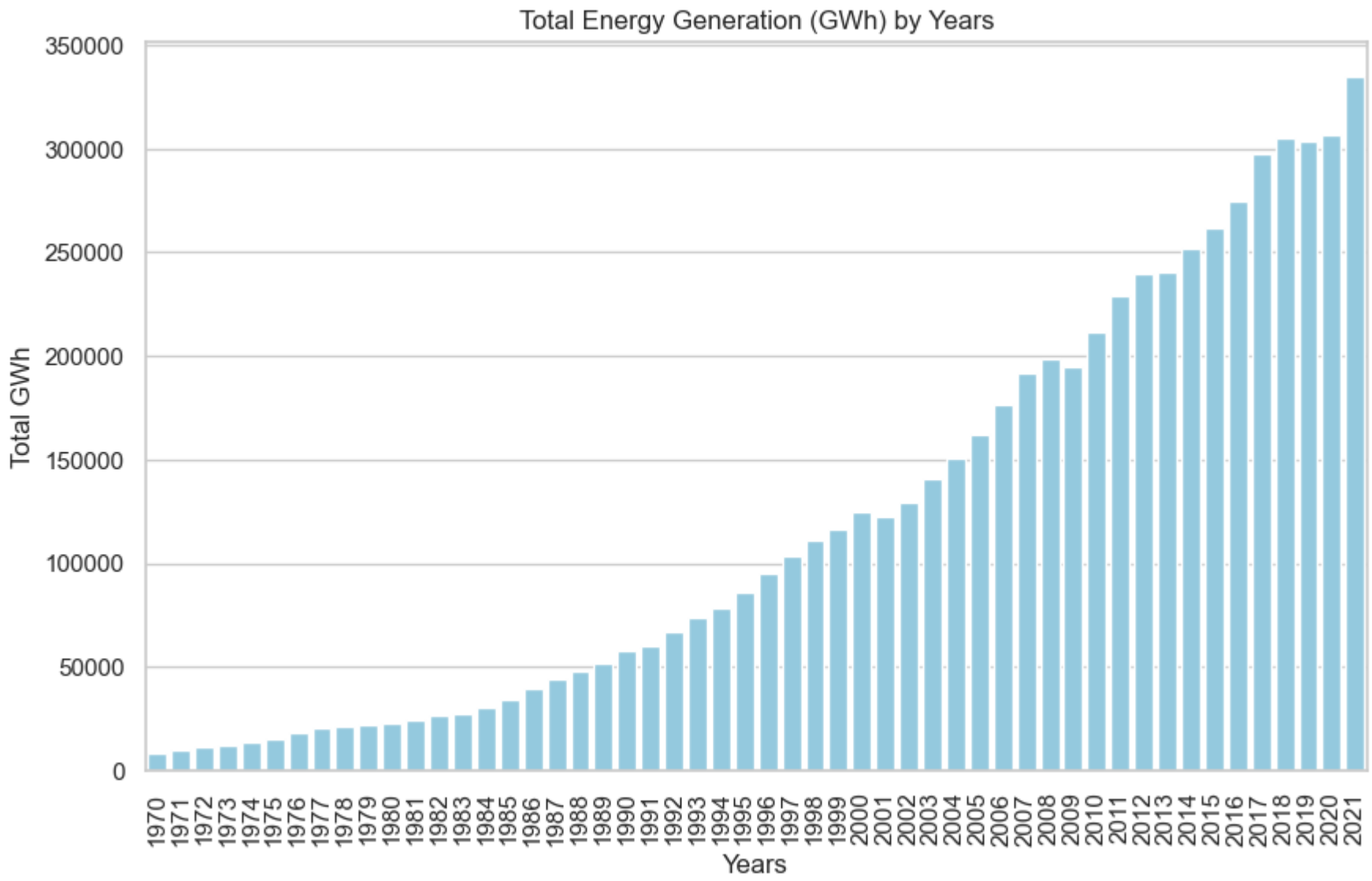
In [41]:
```
df_e_new_numeric = df_e_new.apply(pd.to_numeric, errors='coerce') #Categoric to Numeric
```

In [42]:
```
energy_columns = df_e_new_numeric.columns[2:]

# Years
x = df_e_new_numeric.iloc[:, 0]

sns.set(style="whitegrid")
plt.figure(figsize=(10, 6))

# Line Plot
for column in energy_columns:
    y = df_e_new_numeric[column]
    sns.lineplot(x=x, y=y, label=column)

plt.legend(fontsize='small')

plt.title('Electricity Generation by Energy Resource')
plt.xlabel('Years')
plt.ylabel('Energy Percentage %')

plt.show()
```



- It seems obvious that non-renewable and environmentally harmful methods are mostly preferred in energy production in Turkey.
- The use of liquid fuels in energy production has gradually decreased and is not used at all today.
- The use of renewable energy sources was used little until the 80s, but then they were almost never used for a long time. However, the share of renewable energy sources in energy production has been increasing since 2008. This is a positive development for our country in a world where resources are depleted day by day.

In [43]:
```
x = df_e_new_numeric.iloc[:, 0]
y = df_e_new_numeric.iloc[:, 1]

sns.set(style="whitegrid")

# Line Plot
plt.figure(figsize=(10, 6))
sns.barplot(x=x, y=y, color='skyblue')

plt.xticks(rotation=90)
plt.title('Total Energy Generation (GWh) by Years')
plt.xlabel('Years')
plt.ylabel('Total GWh')

plt.show()
```



It is obvious that Turkey has made serious progress in the field of energy production since 1970.

Greenhouse Gas Emissions

In [44]:
```
df_g.head(10)
```

Out[44]:

	Sera gazı emisyonları (CO2 eşdeğeri), 1990 - 2021	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5
0	Greenhouse gas emissions (CO2 equivalent), 199...	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	(Milyon ton - Million tonnes)
2	Yıl\nYear	Toplam\nTotal	CO2	CH4	N2O	F-gazlar\nF-gases
3	1990	219.526151	151.614981	42.487543	24.950823	0.472804
4	1991	226.794721	158.126315	43.377429	24.733992	0.556985
5	1992	233.132483	164.064876	43.288202	25.25965	0.519754
6	1993	240.771746	171.138875	43.06554	26.048287	0.519045
7	1994	234.38783	167.565647	42.787878	23.570421	0.463884
8	1995	248.248909	181.355257	42.613471	23.870854	0.409326
9	1996	267.581386	199.646693	42.968946	24.54246	0.423287

In [45]:
```
df_g.iloc[3:35]
```

Out[45]:

	Sera gazı emisyonları (CO2 eşdeğeri), 1990 - 2021	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5
3	1990	219.526151	151.614981	42.487543	24.950823	0.472804
4	1991	226.794721	158.126315	43.377429	24.733992	0.556985
5	1992	233.132483	164.064876	43.288202	25.25965	0.519754
6	1993	240.771746	171.138875	43.06554	26.048287	0.519045
7	1994	234.38783	167.565647	42.787878	23.570421	0.463884
8	1995	248.248909	181.355257	42.613471	23.870854	0.409326
9	1996	267.581386	199.646693	42.968946	24.54246	0.423287
10	1997	278.813841	212.108198	42.231314	24.050525	0.423804
11	1998	280.318959	212.135043	42.421875	25.338883	0.423158
12	1999	277.776799	207.927912	43.789483	25.636443	0.422961
13	2000	298.916755	229.93666	43.667289	24.774565	0.538241
14	2001	279.740138	213.581229	42.902486	22.600494	0.655929
15	2002	285.623365	221.225642	40.956329	22.594353	0.847041
16	2003	304.794788	236.765649	42.958062	24.006964	1.064113
17	2004	314.42357	244.768581	43.501239	24.802051	1.351699
18	2005	337.574998	264.870636	45.171945	25.968599	1.563818
19	2006	358.002535	282.368323	46.610152	27.247719	1.776341
20	2007	391.697006	313.693363	49.089652	26.757611	2.15638
21	2008	388.494556	310.622132	50.158793	25.402102	2.311529
22	2009	395.177141	316.412299	49.885842	26.553589	2.325411
23	2010	398.793165	316.193043	51.645279	27.44736	3.507482
24	2011	428.617589	342.057326	54.165558	28.531906	3.862799
25	2012	448.184458	356.107201	57.830017	29.650376	4.596865
26	2013	440.195301	347.331771	56.663033	31.460342	4.740155
27	2014	459.489508	364.028278	58.545059	31.724948	5.191223
28	2015	474.967535	384.92966	52.784799	32.262329	4.990747
29	2016	501.107906	405.95021	55.58199	34.347817	5.227889
30	2017	528.565929	430.900877	56.817086	35.443562	5.404404
31	2018	523.108034	422.05862	60.409228	35.455448	5.184738
32	2019	508.725873	402.692254	63.216502	36.967398	5.84972
33	2020	523.990817	412.926872	63.893758	40.49057	6.679617
34	2021	564.38975	452.70279	64.02023	40.306091	7.36064

In [46]:
```
df_g_new = df_g.iloc[3:35,:] #Clean data
```

In [47]:
```
df_g_new.columns = ['Year','Total','CO2','CH4','N2O','F-Gases']
df_g_new.head()
```

Out[47]:

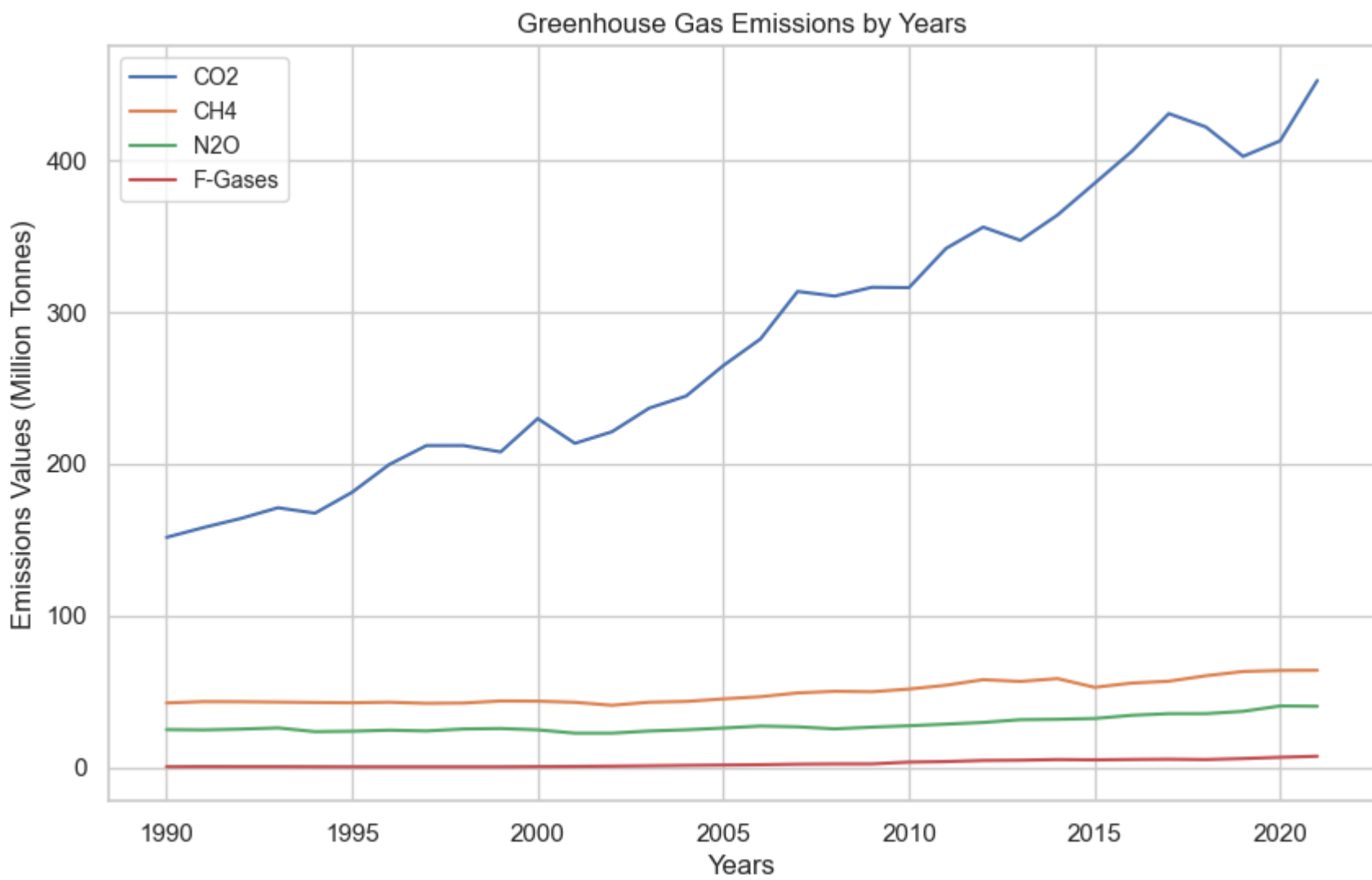
	Year	Total	CO2	CH4	N2O	F-Gases
3	1990	219.526151	151.614981	42.487543	24.950823	0.472804
4	1991	226.794721	158.126315	43.377429	24.733992	0.556985
5	1992	233.132483	164.064876	43.288202	25.25965	0.519754
6	1993	240.771746	171.138875	43.06554	26.048287	0.519045
7	1994	234.38783	167.565647	42.787878	23.570421	0.463884

In [48]:
```
df_g_new.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32 entries, 3 to 34
Data columns (total 6 columns):
 #   Column   Non-Null Count  Dtype 
---  ------   --------------  ----- 
 0   Year     32 non-null     object
 1   Total    32 non-null     object
 2   CO2      32 non-null     object
 3   CH4      32 non-null     object
 4   N2O      32 non-null     object
 5   F-Gases  32 non-null     object
dtypes: object(6)
memory usage: 1.6+ KB
```

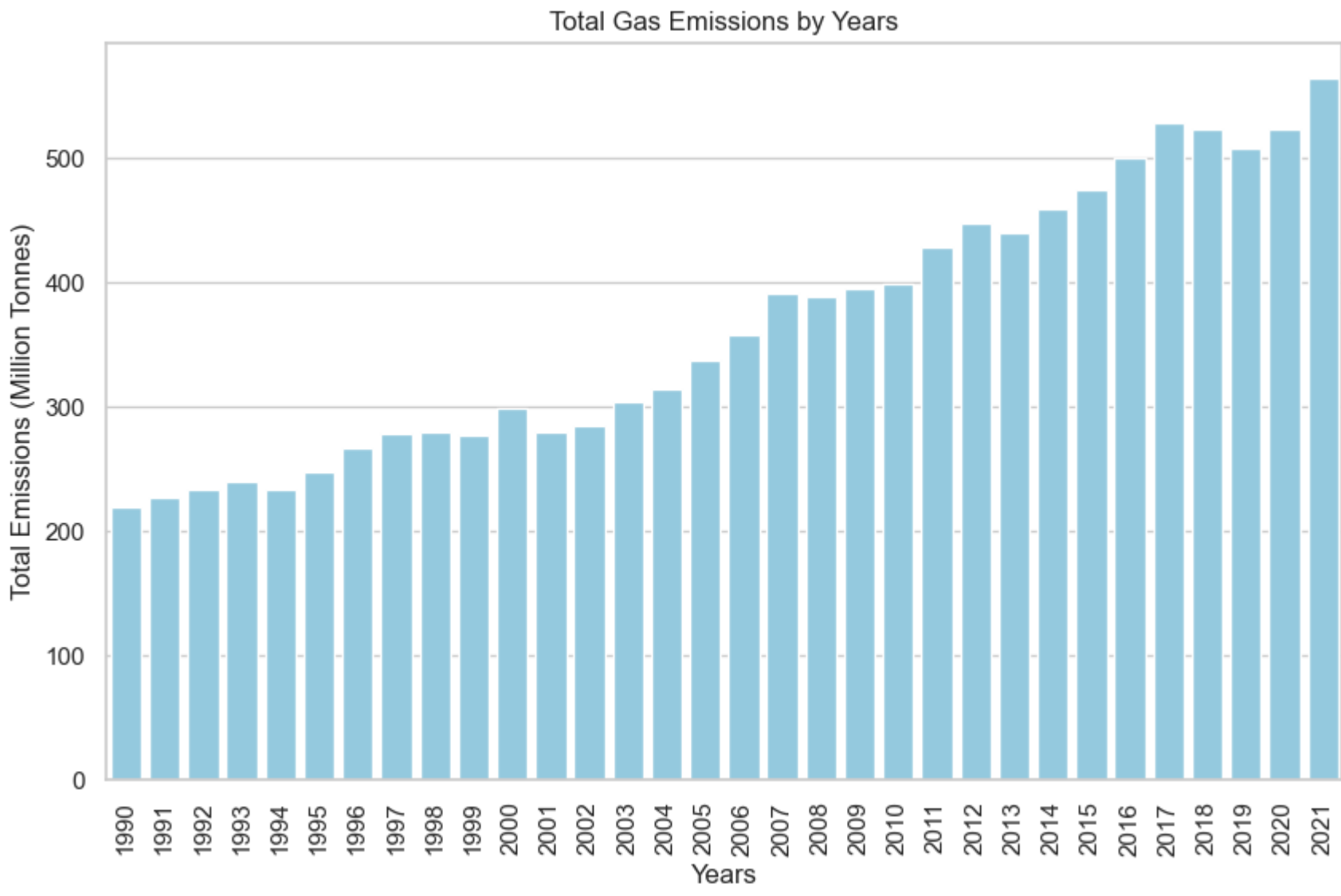
In [49]:
```
df_g_new_numeric = df_g_new.apply(pd.to_numeric, errors='coerce') #Categoric to Numeric
```

In [50]:
```
carbon_columns = df_g_new_numeric.columns[2:]

# Years
x = df_g_new_numeric.iloc[:, 0]

sns.set(style="whitegrid")
plt.figure(figsize=(10, 6))

# Line Plot
for column in carbon_columns:
    y = df_g_new_numeric[column]
    sns.lineplot(x=x, y=y, label=column)

plt.legend(fontsize='small')

plt.title('Greenhouse Gas Emissions by Years')
plt.xlabel('Years')
plt.ylabel('Emissions Values (Million Tonnes)')

plt.show()
```



The graph shows that the emission of CO_2 gas into the atmosphere is increasing day by day, but there is no significant change in the emission of other greenhouse gases. We would not be wrong if we say that our country is facing serious dangers because CO_2 gas is the most important greenhouse gas.

In [51]:
```
x = df_g_new_numeric.iloc[:, 0]
y = df_g_new_numeric.iloc[:, 1]

sns.set(style="whitegrid")

# Line Plot
plt.figure(figsize=(10, 6))
sns.barplot(x=x, y=y, color='skyblue')

plt.xticks(rotation=90)
plt.title('Total Gas Emissions by Years')
plt.xlabel('Years')
plt.ylabel('Total Emissions (Million Tonnes)')

plt.show()
```



In this graph, we can easily observe that total greenhouse gas emissions are increasing over the years almost without any hesitation.

Death Reasons

In [52]:
```
df_d.head(20) # We'll use just number of deaths in respiratory system diseases by years
```

Out[52]:

	Yaş grubu ve cinsiyete göre seçilmiş ölüm nedenlerinin dağılımı, 2009-2017	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8	Unnamed: 9	...	Unnamed: 101	Unnamed: 102
0	Distribution of selected causes of death by ag...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	2009(1)	NaN	NaN	NaN	NaN	...	2017(1)	NaN
2	NaN	NaN	NaN	NaN	NaN	Yaş grubu - Age group	NaN	NaN	NaN	NaN	...	Yaş grubu - Age group	NaN
3	Ölüm nedeni	NaN	NaN	NaN	Toplam	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
4	Cause of death	NaN	NaN	NaN	Total	0 - 14	15 - 24	25 - 34	35 - 44	45 - 54	...	0 - 14	15 - 24
5	Toplam	NaN	NaN	Toplam - Total	281504	17825	3682	5031	9369	22973	...	20087	5839
6	Total	NaN	NaN	Erkek - Male	155928	9768	2412	3241	6020	15661	...	11117	4363
7	NaN	NaN	NaN	Kadın - Female	125576	8057	1270	1790	3349	7312	...	8922	1474
8	NaN	NaN	NaN	Bilinmeyen - Unknown	-	-	-	-	-	-	...	48	2
9	Dolaşım sistemi hastalıkları	NaN	NaN	Toplam - Total	112158	828	336	734	2436	7616	...	546	410
10	Diseases of the circulatory system	NaN	NaN	Erkek - Male	56447	419	183	442	1656	5371	...	264	272
11	NaN	NaN	NaN	Kadın - Female	55711	409	153	292	780	2245	...	282	138
12	NaN	NaN	NaN	Bilinmeyen - Unknown	-	-	-	-	-	-	...	-	-
13	İyi huylu ve kötü huylu tümörler	NaN	NaN	Toplam - Total	59386	726	633	1034	3049	8461	...	715	637
14	(malign ve benign neoplazmlar)	NaN	NaN	Erkek - Male	38736	411	371	542	1638	5526	...	389	388
15	Neoplasms	NaN	NaN	Kadın - Female	20650	315	262	492	1411	2935	...	326	249
16	NaN	NaN	NaN	Bilinmeyen - Unknown	-	-	-	-	-	-	...	-	-
17	Solunum sistemi hastalıkları	NaN	NaN	Toplam - Total	24996	934	185	295	428	1198	...	841	232
18	Diseases of the respiratory system	NaN	NaN	Erkek - Male	15654	485	91	142	256	819	...	463	163
19	NaN	NaN	NaN	Kadın - Female	9342	449	94	153	172	379	...	378	69

20 rows × 111 columns

In [53]:
```
df_d_new = df_d.iloc[17,[4,16,28,40,52,64,76,88,100]].transpose() # Clean data
df_d_new = pd.DataFrame(df_d_new)
df_d_new['Years'] = range(2009, 2018)
df_d_new.index = range(len(df_d_new)) # index
df_d_new = df_d_new.rename(columns={17: 'Number of Deaths'})
df_d_new
```

Out[53]:

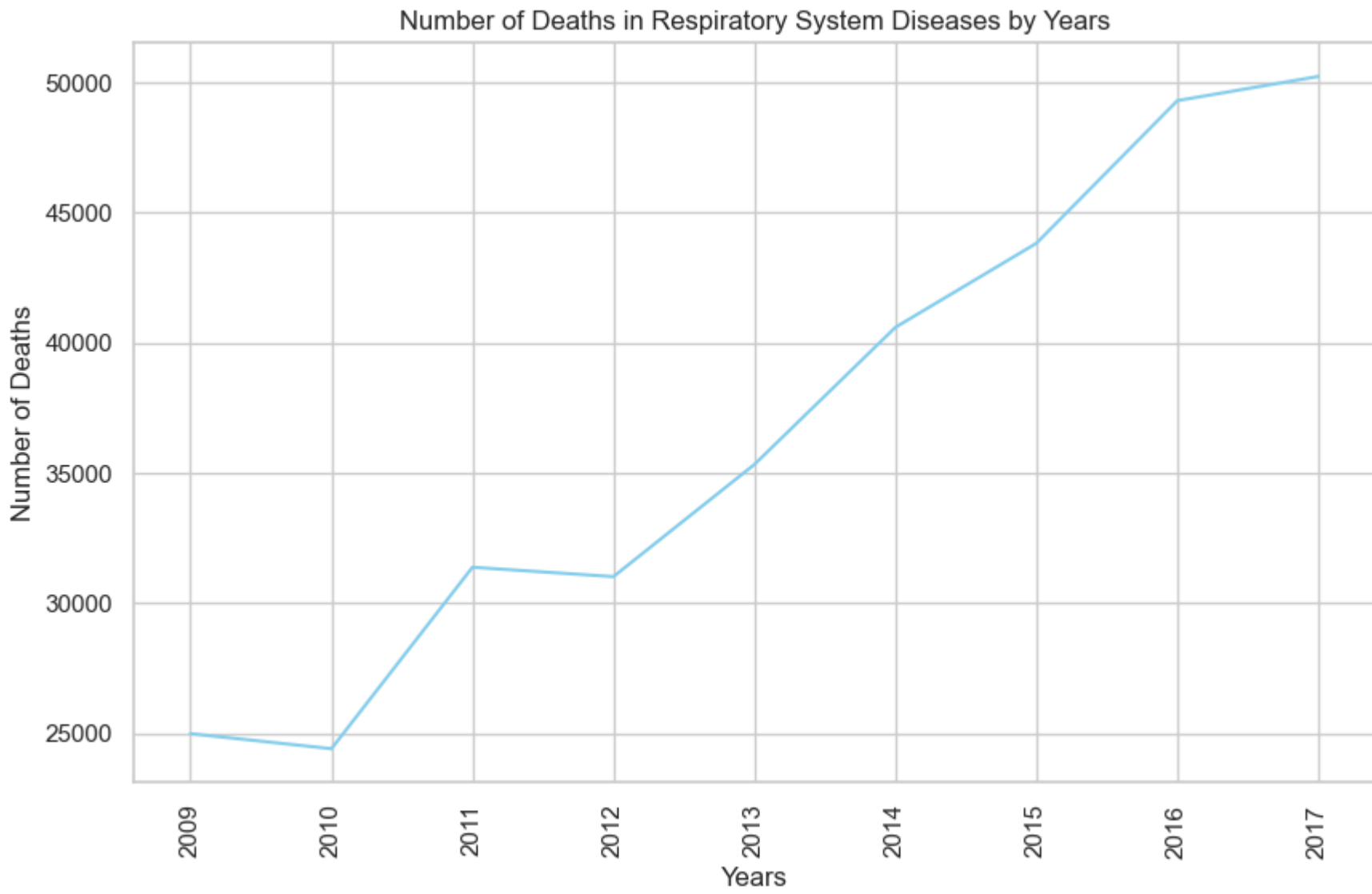
	Number of Deaths	Years
0	24996	2009
1	24418	2010
2	31384	2011
3	31024	2012
4	35331	2013
5	40593	2014
6	43820	2015
7	49296	2016
8	50224	2017

In [54]:
```
df_d_new.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 2 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Number of Deaths  9 non-null      object
 1   Years             9 non-null      int64
dtypes: int64(1), object(1)
memory usage: 276.0+ bytes
```

In [55]:
```
df_d_new_numeric = df_d_new.apply(pd.to_numeric, errors='coerce') #Categoric to Numeric
```

In [56]:
```
y = df_d_new_numeric.iloc[:, 0]
x = df_d_new_numeric.iloc[:, 1]

sns.set(style="whitegrid")

# Line Plot
plt.figure(figsize=(10, 6))
sns.lineplot(x=x, y=y, color='skyblue')

plt.xticks(rotation=90)
plt.title('Number of Deaths in Respiratory System Diseases by Years')
plt.xlabel('Years')
plt.ylabel('Number of Deaths')

plt.show()
```



When we look at the graph in general, we see that deaths due to respiratory diseases have doubled in our country in just 8 years. These types of deaths have been increasing without any hesitation, especially since 2012. This situation literally screams that there is something wrong in our country regarding the respiratory systems of our citizens. The aim of this project is to analyze the causes of such deaths in the context of energy production methods and greenhouse gas emissions and to prevent deaths.

Research Questions and Proposed Tests

1st Question

Is there a significant relationship between greenhouse gas emissions and deaths from respiratory diseases?

Hypothesis: As greenhouse gas emissions increase, deaths due to respiratory diseases will increase.

In [57]:
```
df_g_new_numeric
```

Out[57]:

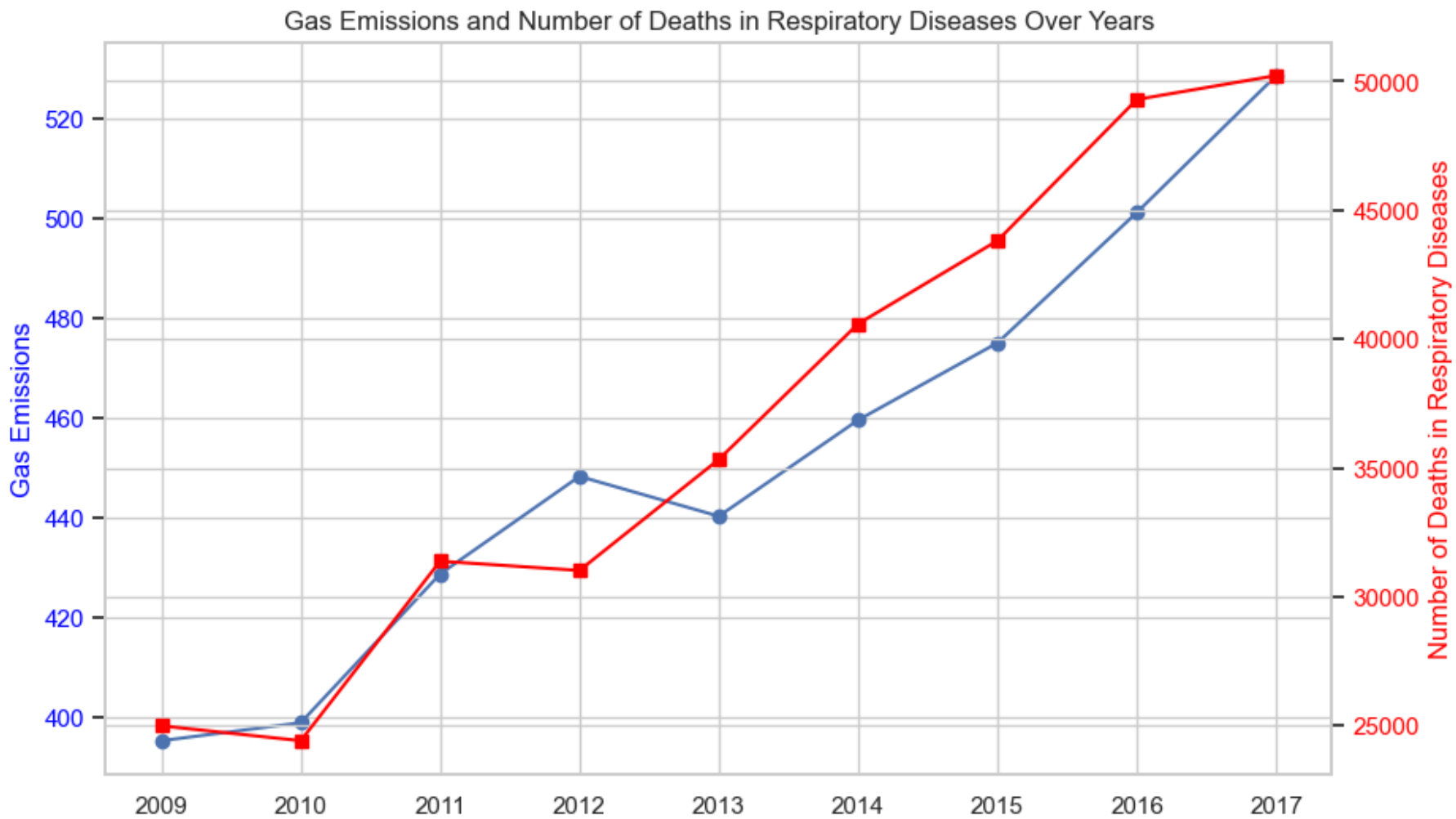
	Year	Total	CO2	CH4	N2O	F-Gases
3	1990	219.526151	151.614981	42.487543	24.950823	0.472804
4	1991	226.794721	158.126315	43.377429	24.733992	0.556985
5	1992	233.132483	164.064876	43.288202	25.259650	0.519754
6	1993	240.771746	171.138875	43.065540	26.048287	0.519045
7	1994	234.387830	167.565647	42.787878	23.570421	0.463884
8	1995	248.248909	181.355257	42.613471	23.870854	0.409326
9	1996	267.581386	199.646693	42.968946	24.542460	0.423287
10	1997	278.813841	212.108198	42.231314	24.050525	0.423804
11	1998	280.318959	212.135043	42.421875	25.338883	0.423158
12	1999	277.776799	207.927912	43.789483	25.636443	0.422961
13	2000	298.916755	229.936660	43.667289	24.774565	0.538241
14	2001	279.740138	213.581229	42.902486	22.600494	0.655929
15	2002	285.623365	221.225642	40.956329	22.594353	0.847041
16	2003	304.794788	236.765649	42.958062	24.006964	1.064113
17	2004	314.423570	244.768581	43.501239	24.802051	1.351699
18	2005	337.574998	264.870636	45.171945	25.968599	1.563818
19	2006	358.002535	282.368323	46.610152	27.247719	1.776341
20	2007	391.697006	313.693363	49.089652	26.757611	2.156380
21	2008	388.494556	310.622132	50.158793	25.402102	2.311529
22	2009	395.177141	316.412299	49.885842	26.553589	2.325411
23	2010	398.793165	316.193043	51.645279	27.447360	3.507482
24	2011	428.617589	342.057326	54.165558	28.531906	3.862799
25	2012	448.184458	356.107201	57.830017	29.650376	4.596865
26	2013	440.195301	347.331771	56.663033	31.460342	4.740155
27	2014	459.489508	364.028278	58.545059	31.724948	5.191223
28	2015	474.967535	384.929660	52.784799	32.262329	4.990747
29	2016	501.107906	405.950210	55.581990	34.347817	5.227889
30	2017	528.565929	430.900877	56.817086	35.443562	5.404404
31	2018	523.108034	422.058620	60.409228	35.455448	5.184738
32	2019	508.725873	402.692254	63.216502	36.967398	5.849720
33	2020	523.990817	412.926872	63.893758	40.490570	6.679617
34	2021	564.389750	452.702790	64.020230	40.306091	7.360640

In [58]:
```
df_corr= pd.DataFrame()
df_corr['Years']=df_g_new_numeric.iloc[19:28,0]
df_corr['Gas Emissions']=df_g_new_numeric.iloc[19:28,1]
df_corr=df_corr.reset_index().drop(['index'],axis=1)
df_corr['#Deaths in Respiratory Diseases']=df_d_new_numeric['Number of Deaths']
df_corr
```

Out[58]:

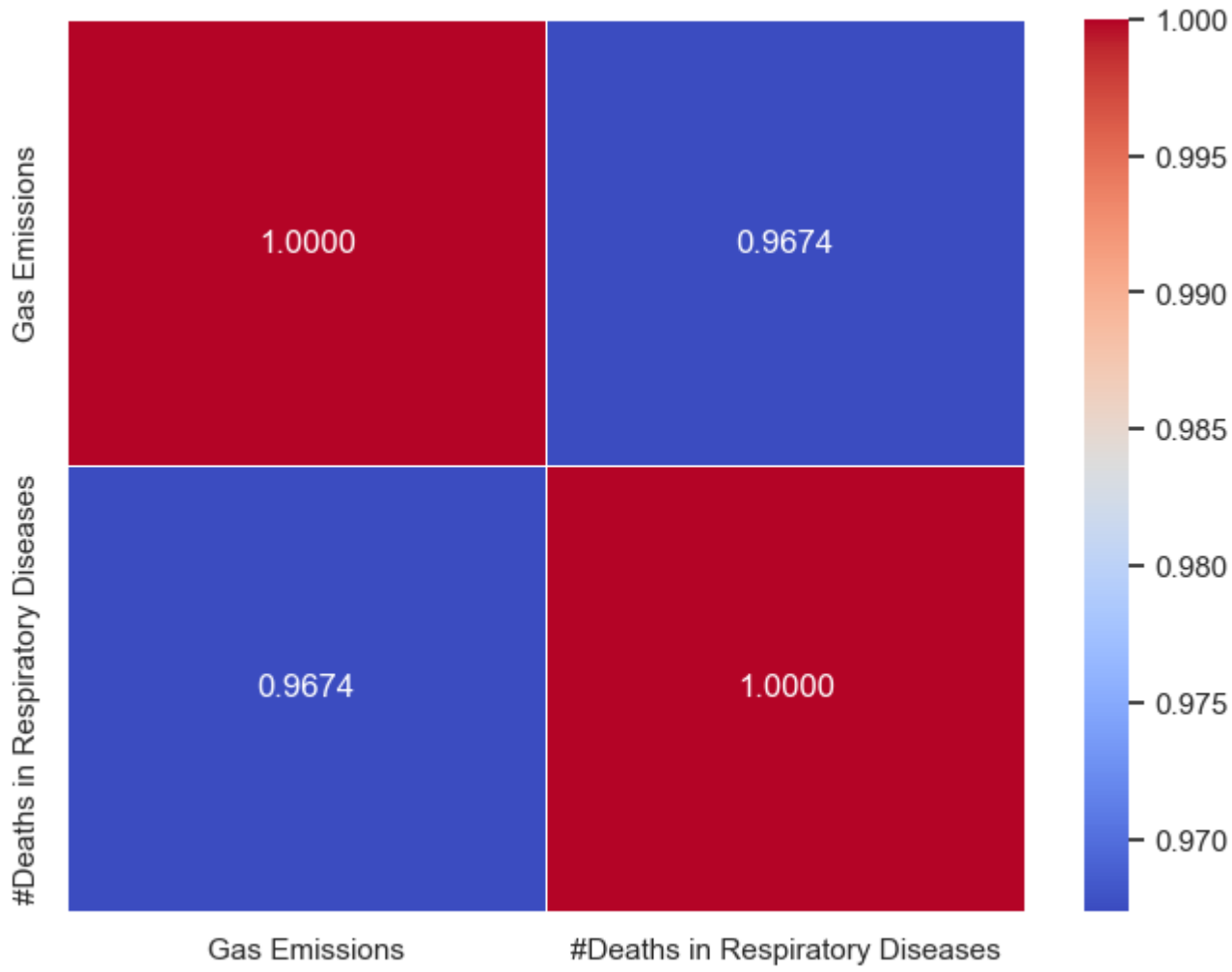
	Years	Gas Emissions	#Deaths in Respiratory Diseases
0	2009	395.177141	24996
1	2010	398.793165	24418
2	2011	428.617589	31384
3	2012	448.184458	31024
4	2013	440.195301	35331
5	2014	459.489508	40593
6	2015	474.967535	43820
7	2016	501.107906	49296
8	2017	528.565929	50224

In [59]:
```
plt.figure(figsize=(10, 6))
plt.plot(df_corr['Years'], df_corr['Gas Emissions'], label='Gas Emissions', marker='o')

plt.ylabel('Gas Emissions', color='blue')
plt.tick_params(axis='y', labelcolor='blue')

plt.twinx()
plt.plot(df_corr['Years'], df_corr['#Deaths in Respiratory Diseases'], label='#Deaths in Respiratory Diseases',
         marker='s', color='red')
plt.ylabel('Number of Deaths in Respiratory Diseases', color='red')
plt.tick_params(axis='y', labelcolor='red')

plt.title('Gas Emissions and Number of Deaths in Respiratory Diseases Over Years')
plt.xlabel('Years')
plt.show()
```



It can be seen that greenhouse gas emissions and the number of deaths due to respiratory diseases between 2009 and 2017 draw similar graphs.

In [60]:
```
correlation_matrix = df_corr[['Gas Emissions', '#Deaths in Respiratory Diseases']].corr()

# Heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".4f", linewidths=.5)
plt.show()
```



The incredibly high correlation value of approximately 97% completely confirms the previous graph. In other words, greenhouse gas emissions seriously affect people's respiratory system health.

2nd Question

How does energy production method relate to the annual number of respiratory deaths?

Hypothesis: While non-environmentally friendly energy sources have an effect on increasing the number of deaths due to respiratory diseases, environmentally friendly energy sources do not have this effect.

In [61]:
```
df_e_new_numeric['Harmful Sources%'] = df_e_new_numeric[['Coal%', 'Liquid Fuels%', 'Natural Gas%']].sum(axis=1)

df_e_new_numeric['Product by Harmful Sources'] = (df_e_new_numeric['Harmful Sources%'] * df_e_new_numeric['Total (GWh)
df_e_new_numeric
```

Out[61]:

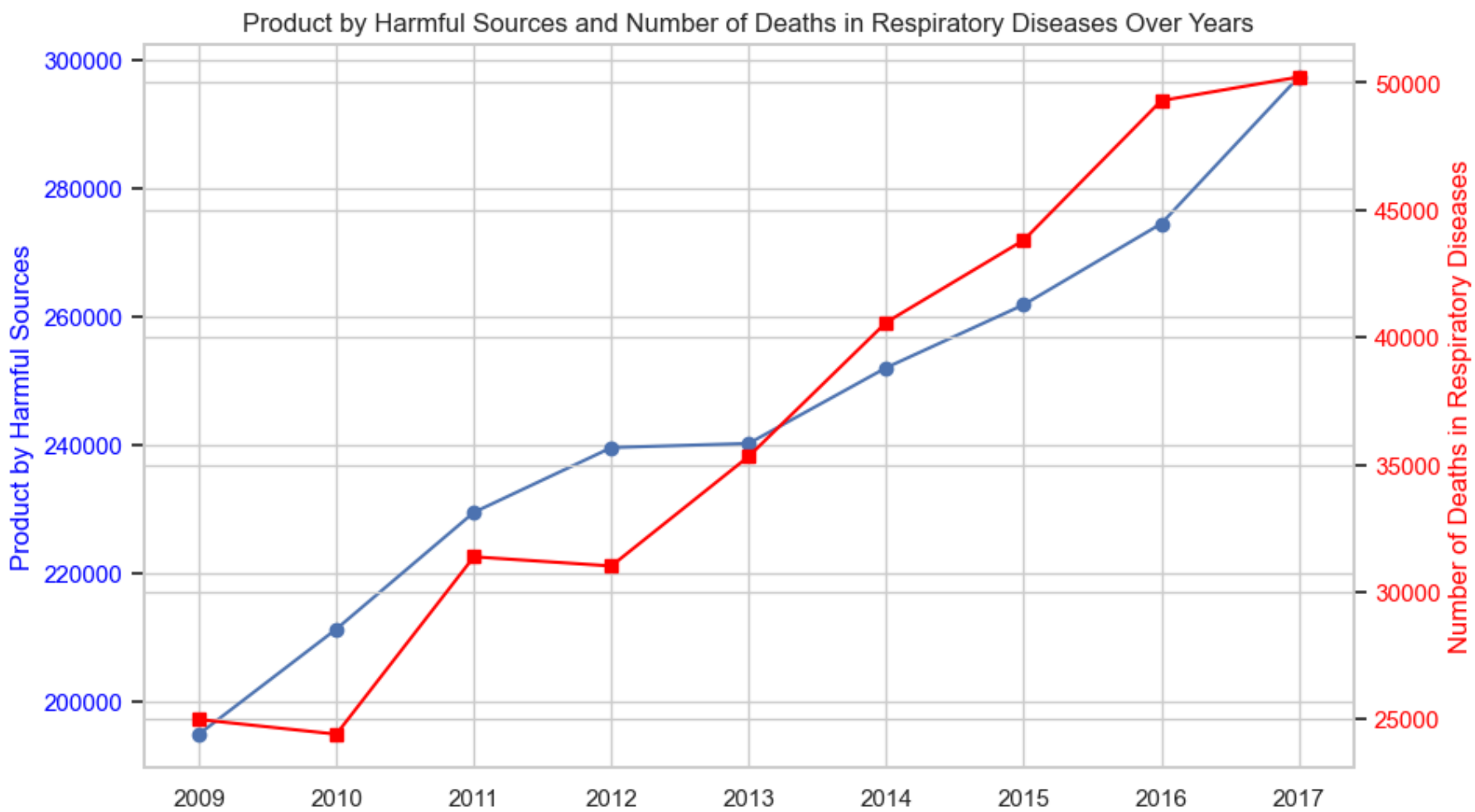
	Year	Total (GWh)	Coal%	Liquid Fuels%	Natural Gas%	Hydro%	Renewable Energy and Wastes%	Harmful Sources%	Product by Harmful Sources
4	1970	8623.000000	32.700000	30.200000	NaN	35.200000	1.900000	62.900000	5423.867000
5	1971	9781.000000	30.400000	41.200000	NaN	26.700000	1.700000	71.600000	7003.196000
6	1972	11242.000000	26.000000	43.900000	NaN	28.500000	1.600000	69.900000	7858.158000
7	1973	12425.000000	26.100000	51.300000	NaN	21.000000	1.600000	77.400000	9616.950000
8	1974	13477.000000	28.800000	44.800000	NaN	24.900000	1.500000	73.600000	9919.072000
9	1975	15623.000000	26.300000	34.500000	NaN	37.800000	1.400000	60.800000	9498.784000
10	1976	18283.000000	23.700000	29.600000	NaN	45.800000	0.900000	53.300000	9744.839000
11	1977	20565.000000	23.800000	33.400000	NaN	41.700000	1.100000	57.200000	11763.180000
12	1978	21726.000000	25.700000	30.700000	NaN	43.000000	0.600000	56.400000	12253.464000
13	1979	22522.000000	28.600000	25.100000	NaN	45.700000	0.600000	53.700000	12094.314000
14	1980	23275.000000	25.600000	25.000000	NaN	48.800000	0.600000	50.600000	11777.150000
15	1981	24673.000000	24.900000	23.600000	NaN	51.100000	0.400000	48.500000	11966.405000
16	1982	26552.000000	24.200000	22.400000	NaN	53.400000	0.000000	46.600000	12373.232000
17	1983	27347.000000	31.400000	27.100000	NaN	41.500000	0.000000	58.500000	15997.995000
18	1984	30614.000000	33.000000	23.000000	NaN	43.900000	0.100000	56.000000	17143.840000
19	1985	34219.000000	43.900000	20.700000	0.200000	35.200000	0.000000	64.800000	22173.912000
20	1986	39695.000000	49.000000	17.600000	3.400000	29.900000	0.100000	70.000000	27786.500000
21	1987	44353.000000	39.800000	12.400000	5.700000	42.000000	0.100000	57.900000	25680.387000
22	1988	48049.000000	26.000000	6.900000	6.700000	60.300000	0.100000	39.600000	19027.404000
23	1989	52043.000000	38.900000	8.200000	18.300000	34.500000	0.100000	65.400000	34036.122000
24	1990	57543.000000	35.100000	6.800000	17.700000	40.200000	0.200000	59.600000	34295.628000
25	1991	60246.000000	35.800000	5.600000	20.800000	37.600000	0.200000	62.200000	37473.012000
26	1992	67342.000000	36.500000	7.800000	16.000000	39.500000	0.200000	60.300000	40607.226000
27	1993	73808.000000	32.100000	7.000000	14.600000	46.100000	0.200000	53.700000	39634.896000
28	1994	78322.000000	36.000000	7.100000	17.600000	39.100000	0.200000	60.700000	47541.454000
29	1995	86247.000000	32.500000	6.700000	19.200000	41.200000	0.400000	58.400000	50368.248000
30	1996	94862.000000	32.000000	6.900000	18.100000	42.700000	0.300000	57.000000	54071.340000
31	1997	103296.000000	32.800000	6.900000	21.400000	38.500000	0.400000	61.100000	63113.856000
32	1998	111022.000000	32.200000	7.200000	22.400000	38.000000	0.300000	61.800000	68611.596000
33	1999	116440.000000	31.800000	6.900000	31.200000	29.800000	0.300000	69.900000	81391.560000
34	2000	124922.000000	30.600000	7.500000	37.000000	24.700000	0.300000	75.100000	93816.422000
35	2001	122725.000000	31.300000	8.400000	40.400000	19.600000	0.300000	80.100000	98302.725000
36	2002	129400.000000	24.800000	8.300000	40.600000	26.000000	0.300000	73.700000	95367.800000
37	2003	140581.000000	22.900000	6.600000	45.200000	25.100000	0.200000	74.700000	105014.007000
38	2004	150698.000000	22.800000	5.000000	41.300000	30.600000	0.300000	69.100000	104132.318000
39	2005	161956.000000	26.600000	3.400000	45.300000	24.400000	0.300000	75.300000	121952.868000
40	2006	176300.000000	26.400000	2.400000	45.800000	25.100000	0.300000	74.600000	131519.800000
41	2007	191558.000000	27.900000	3.400000	49.600000	18.700000	0.400000	80.900000	154970.422000
42	2008	198418.000000	29.100000	3.800000	49.700000	16.800000	0.600000	82.600000	163893.268000
43	2009	194813.000000	28.600000	2.500000	49.300000	18.500000	1.200000	80.400000	156629.652000
44	2010	211208.000000	26.100000	1.000000	46.500000	24.500000	1.900000	73.600000	155449.088000
45	2011	229395.000000	28.800000	0.400000	45.400000	22.800000	2.600000	74.600000	171128.670000
46	2012	239497.000000	28.400000	0.700000	43.600000	24.200000	3.100000	72.700000	174114.319000
47	2013	240154.000000	26.600000	0.700000	43.800000	24.700000	4.200000	71.100000	170749.494000
48	2014	251963.000000	30.200000	0.900000	47.900000	16.100000	4.900000	79.000000	199050.770000
49	2015	261783.000000	29.100000	0.900000	37.900000	25.600000	6.500000	67.900000	177750.657000
50	2016	274407.700000	33.700000	0.700000	32.500000	24.500000	8.600000	66.900000	183578.751300
51	2017	297277.500000	32.800000	0.400000	37.200000	19.600000	10.000000	70.400000	209283.360000
52	2018	304802.000000	37.200000	0.100000	30.300000	19.700000	12.700000	67.600000	206046.152000
53	2019	303897.600000	37.148743	0.110565	18.851155	29.227867	14.661670	56.110463	170518.350659
54	2020	306703.092444	34.499816	0.105203	23.127036	25.462531	16.805415	57.732055	177065.997336
55	2021	334723.111929	30.885850	0.084096	33.215739	16.708380	19.105935	64.185685	214844.322520

In [62]:
```
df_= pd.DataFrame()
df_['Years']=df_e_new_numeric.iloc[39:48,0]
df_['Product by Harmful Sources']=df_e_new_numeric.iloc[39:48,1]
df_=df_.reset_index().drop(['index'],axis=1)
df_['#Deaths in Respiratory Diseases']=df_d_new_numeric['Number of Deaths']
df_
```

Out[62]:

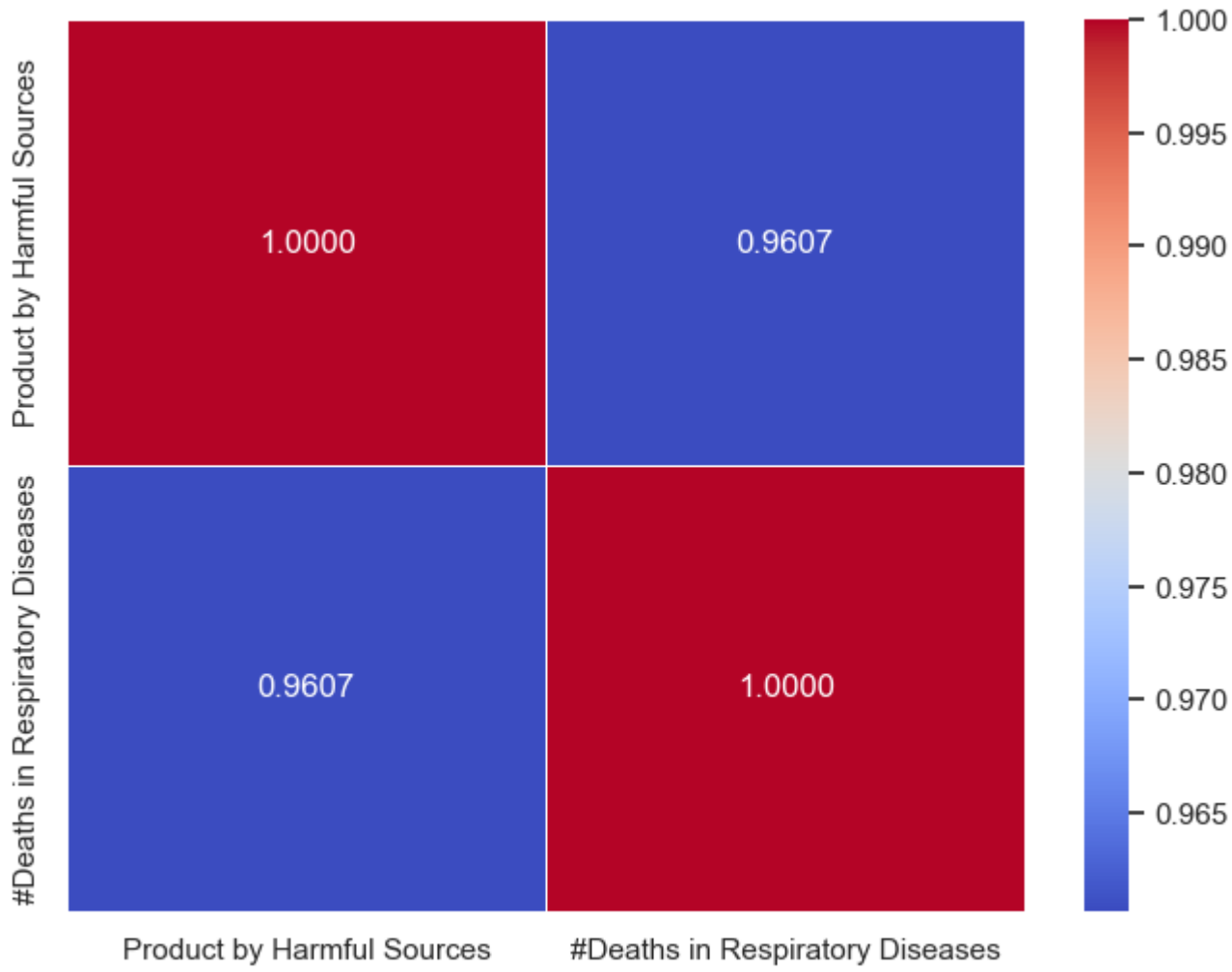
	Years	Product by Harmful Sources	#Deaths in Respiratory Diseases
0	2009	194813.0	24996
1	2010	211208.0	24418
2	2011	229395.0	31384
3	2012	239497.0	31024
4	2013	240154.0	35331
5	2014	251963.0	40593
6	2015	261783.0	43820
7	2016	274407.7	49296
8	2017	297277.5	50224

In [63]:
```
plt.figure(figsize=(10, 6))
plt.plot(df_['Years'], df_['Product by Harmful Sources'], label='Product by Harmful Sources', marker='o')

plt.ylabel('Product by Harmful Sources', color='blue')
plt.tick_params(axis='y', labelcolor='blue')

plt.twinx()
plt.plot(df_['Years'], df_['#Deaths in Respiratory Diseases'], label='#Deaths in Respiratory Diseases', marker='s',
         color='red')
plt.ylabel('Number of Deaths in Respiratory Diseases', color='red')
plt.tick_params(axis='y', labelcolor='red')

plt.title('Product by Harmful Sources and Number of Deaths in Respiratory Diseases Over Years')
plt.xlabel('Years')
plt.show()
```



It is seen that deaths due to respiratory diseases have also increased with the increase in the use of environmentally harmful resources in energy production within the specified time period. These sources directly or indirectly cause an increase in greenhouse gas emissions and harm the health of the respiratory system.

In [64]:
```
correlation_matrix = df_[['Product by Harmful Sources', '#Deaths in Respiratory Diseases']].corr()

# Heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".4f", linewidths=.5)
plt.show()
```



3rd Question

What is the relationship between the energy production method and the amount of greenhouse gases in the atmosphere?

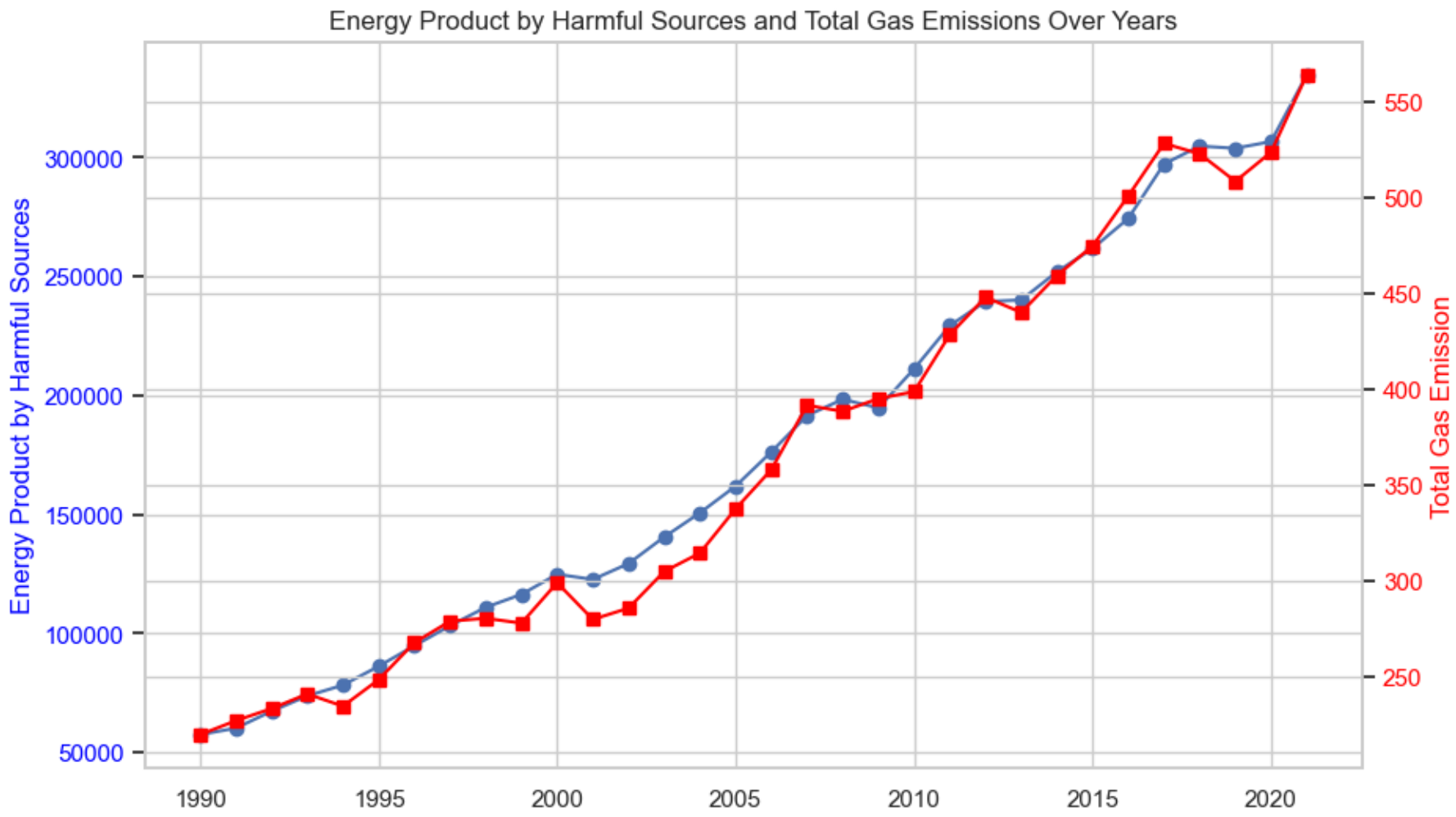
Hypothesis: The use of non-environmentally friendly, non-renewable energy sources increases greenhouse gas emissions.

In [65]:
```
dff= pd.DataFrame()
dff['Years']=df_e_new_numeric.iloc[20:52,0]
dff['Product by Harmful Sources']=df_e_new_numeric.iloc[20:52,1]
dff=dff.reset_index().drop(['index'],axis=1)
df_g_new_numeric=df_g_new_numeric.reset_index().drop(['index'],axis=1)
dff['Total Gas Emission']=df_g_new_numeric.iloc[:,1]
dff
```

Out[65]:

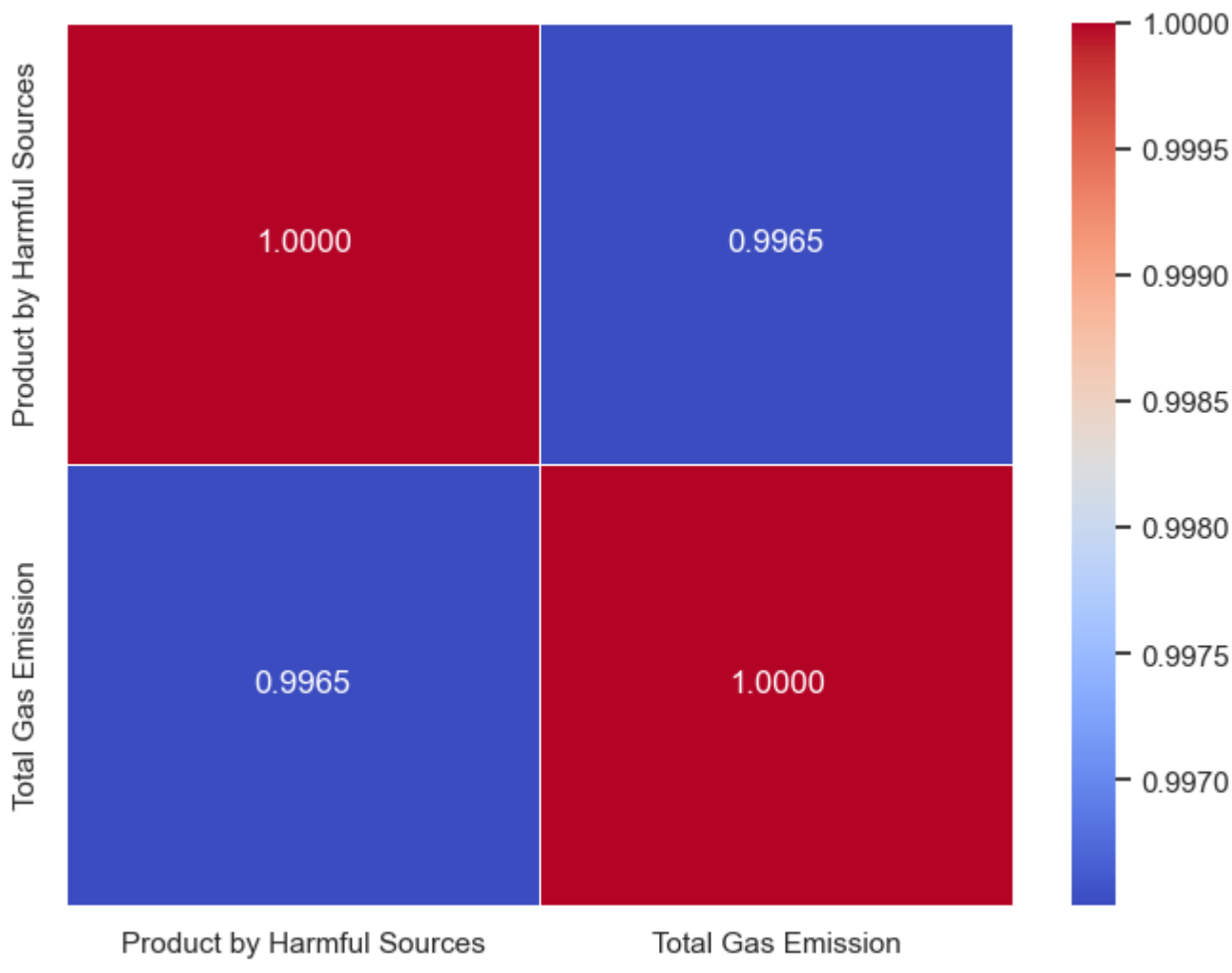
	Years	Product by Harmful Sources	Total Gas Emission
0	1990	57543.000000	219.526151
1	1991	60246.000000	226.794721
2	1992	67342.000000	233.132483
3	1993	73808.000000	240.771746
4	1994	78322.000000	234.387830
5	1995	86247.000000	248.248909
6	1996	94862.000000	267.581386
7	1997	103296.000000	278.813841
8	1998	111022.000000	280.318959
9	1999	116440.000000	277.776799
10	2000	124922.000000	298.916755
11	2001	122725.000000	279.740138
12	2002	129400.000000	285.623365
13	2003	140581.000000	304.794788
14	2004	150698.000000	314.423570
15	2005	161956.000000	337.574998
16	2006	176300.000000	358.002535
17	2007	191558.000000	391.697006
18	2008	198418.000000	388.494556
19	2009	194813.000000	395.177141
20	2010	211208.000000	398.793165
21	2011	229395.000000	428.617589
22	2012	239497.000000	448.184458
23	2013	240154.000000	440.195301
24	2014	251963.000000	459.489508
25	2015	261783.000000	474.967535
26	2016	274407.700000	501.107906
27	2017	297277.500000	528.565929
28	2018	304802.000000	523.108034
29	2019	303897.600000	508.725873
30	2020	306703.092444	523.990817
31	2021	334723.111929	564.389750

In [66]:
```
plt.figure(figsize=(10, 6))
plt.plot(dff['Years'], dff['Product by Harmful Sources'], label='Product by Harmful Sources', marker='o')

plt.ylabel('Energy Product by Harmful Sources', color='blue')
plt.tick_params(axis='y', labelcolor='blue')

plt.twinx()
plt.plot(dff['Years'], dff['Total Gas Emission'], label='Total Gas Emission', marker='s', color='red')
plt.ylabel('Total Gas Emission', color='red')
plt.tick_params(axis='y', labelcolor='red')

plt.title('Energy Product by Harmful Sources and Total Gas Emissions Over Years')
plt.xlabel('Years')
plt.show()
```



The line graph formed by the amount of energy produced from harmful sources and the greenhouse gas released into the atmosphere is almost parallel to each other. This supports our hypothesis.

In [67]:
```
correlation_matrix = dff[['Product by Harmful Sources', 'Total Gas Emission']].corr()

# Heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".4f", linewidths=.5)
plt.show()
```



These two features show an almost 100% correlation, proving that energy production sources have a very important role in greenhouse gas emissions.

4th Question

In the light of available data, what will be the estimated number of deaths due to respiratory diseases in the coming years?

Hypothesis: Since Turkey produces more energy every year, if the primary energy production sources are not determined as environmentally friendly sources, deaths due to respiratory diseases will continue to increase.

After creating a complete data set by combining all data sets, the number of deaths due to respiratory diseases can be predicted for the following years by training a linear regression model. If current energy production policies are continued in a similar way, the policy to be followed can be optimized by having a prediction about how the health status of citizens will be affected.

In [ ]: